

1. Administrative & Study Identification

Full Study Title: A Study of Sotatercept (MK-7962) in Healthy Females Administered as a Liquid Formulation in an Autoinjector Versus the Lyophilized Formulation (MK-7962-027) **Short Title/ Acronym:** MK-7962-027 **Protocol Number:** MK-7962-027 **NCT Number:** NCT06930664 **Sponsor Name:** Merck **Company:** Merck **Investigational Product:** Sotatercept (MK-7962 / ActRIIA-IgG1Fc) Liquid Formulation (Autoinjector) and Lyophilized Formulation **Preferred UMLS Name:** Sotatercept **Phase of Development:** PHASE1 **Medical Monitor:** Merck Clinical Investigator

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetic parameters of different forms of sotatercept medications in a healthy person's body over time. **Secondary Objectives:** To evaluate the overall safety, tolerability, and other pharmacokinetic properties of sotatercept when administered as a liquid formulation via autoinjector compared to a lyophilized formulation.

2.2 Background & Rationale The goal of the study is to learn what happens to different forms of sotatercept medications in a healthy person's body over time. Researchers want to know if there is a difference in the healthy person's body when different forms of sotatercept medications are given.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Lyophilized Formulation) **Trial Status:** ACTIVE_NOT_RECRUITING **Blinding:** Open-label / Investigator-blinded (Standard for Phase 1 PK Formulation studies) **Randomization:** Randomized **Disease Areas:** Healthy

3.2 Planned Enrollment & Duration Targeted Enrollment: 146 healthy female volunteers **Number of Sites:** 4 locations in the United States **Estimated Study Start:** 16-May-2025 **Estimated Primary Completion:** 11-Nov-2025 (or 30-Jan-2026 based on final clinical tracking)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria The key inclusion criteria include but are not limited to the following: 1. Is in good health before randomization. 2. Has a body weight between 75 and 100 kg, inclusive. 3. Is assigned female sex at birth and is a participant of nonchildbearing potential (PONCBP) defined as one of the following: * Premenarchal * Premenopausal with documented hysterectomy and/or bilateral salpingectomy * Postmenopausal * Medical cause of permanent infertility (e.g., Müllerian agenesis, androgen insensitivity)

4.2 Exclusion Criteria The key exclusion criteria include but are not limited to the following: 1. Has a history of opportunistic infection (e.g., invasive candidiasis or pneumocystis pneumonia). 2. Has had a serious local infection (e.g., cellulitis, abscess) or systemic infection (e.g., septicemia) within 3 months prior to screening. 3. Any condition that might confound the ability to interpret the pharmacokinetic data.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants will receive one of the following single-dose regimens: * A single subcutaneous low dose of sotatercept administered via autoinjector. * A single subcutaneous high dose of sotatercept administered via autoinjector. * A single subcutaneous high dose of sotatercept administered via syringe (lyophilized). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Single administration with subsequent follow-up period for PK blood sampling.

5.2 Concomitant & Prohibited Therapy Permitted: Routine medications not interfering with study drug pharmacokinetics (per investigator discretion). **Prohibited:** Use of other investigational drugs or biologics, and any medications known to interact with sotatercept absorption or clearance.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -1 Informed Consent, Medical History, Physical Exam, Eligibility Labs Baseline / Dosing
Day 1	Randomization, Single Subcutaneous Injection (Autoinjector or Syringe), Initial PK Blood Draws	Follow-up PK	Days 2 to [X] Serial Blood Samples for PK, Safety Monitoring End of Study
Follow-up	Final Vital Signs, Final PK/Safety Labs, AE Tracking		

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Area Under the Concentration-time Curve from Time 0 Extrapolated to Infinity (AUC_{0-inf}) of Sotatercept at High Dose:** Blood samples will be collected to determine the AUC_{0-inf} of sotatercept in plasma.
- 2. Area Under the Concentration-Time Curve from Time 0 to the Time of the Last Quantifiable Concentration (AUC_{0-last}) of Sotatercept at High Dose:** Blood samples will be collected to determine the AUC_{0-last} of sotatercept in plasma.
- 3. Maximum Observed Concentration (C_{max}) of Sotatercept at High Dose:** Blood samples will be collected to determine the C_{max} of sotatercept in plasma.

7.2 Secondary Outcomes

- 1. Time of the Maximum Observed Concentration (T_{max}) of Sotatercept High Dose.**
- 2. Apparent Terminal Elimination Half-Life (t_{1/2}) of Sotatercept at High Dose.**
- 3. Apparent Total Plasma Clearance (CL/F) of Sotatercept at High Dose.**

(Additional exploratory outcomes include incidence and severity of Treatment-Emergent Adverse Events (TEAEs) and injection site reactions associated with the autoinjector versus the syringe).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: An AE is recorded as any untoward medical occurrence in a clinical study participant, temporally associated with the use of the study intervention, whether or not considered related to the study intervention.

Serious Adverse Events (SAEs): Must be reported to the sponsor and regulatory authorities within 24 hours of investigator awareness.

Data Monitoring Committee (DMC): Internal safety review by Merck.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations:

- * *PK Analysis Set:* All randomized subjects who receive the dose and provide evaluable PK data.
- * *Safety Set:* All subjects who receive at least one dose of the investigational product.

Sample Size Justification: A sample size of $N=146$ was calculated to provide adequate power ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to establish bioequivalence or pharmacokinetic comparability between the liquid and lyophilized formulations.

Handling of Missing Data: Standard non-compartmental analysis (NCA) conventions applied for missing PK timepoints; LOCF or multiple imputation for missing safety data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the ethical principles

that have their origin in the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring for source data verification (SDV) of critical PK and safety data. **Audit Trail:** Secure Electronic Data Capture (EDC) system with full audit trails for eCRFs.

11. Study Identifiers & Governance

PostingID: 643150724159281484 **Primary ID:** MK-7962-027 **Registry Number:** NCT06930664 **Sponsor/Lead Organization:** Merck **Company:** Merck **Study Title:** A Study of Sotatercept (MK-7962) in Healthy Females Administered as a Liquid Formulation in an Autoinjector Versus the Lyophilized Formulation (MK-7962-027) **Official Regulatory Title:** A Phase 1 Clinical Study in Healthy Females of Nonchildbearing Potential to Compare the Pharmacokinetics of Sotatercept (MK-7962) Administered as a Liquid Formulation in an Autoinjector Versus the Lyophilized Formulation

12. Clinical Framework (Sotatercept Specific)

Primary Condition: Healthy **Diagnosis Threshold:** N/A (Healthy participants) **Medical Methodology:** Formulation Comparability Assessment **Optimization Target:** Bioavailability and PK equivalence of the autoinjector device vs. standard syringe

13. Study Procedures & Methodology

Management Pathway: Phase 1 Unit Standard Operating Procedures **Education Protocol (HCP):** Site Initiation Visit (SIV) and autoinjector device usage training **Education Protocol (Patient):** Informed consent process detailing single-dose SC administration and PK schedule **Quality Audit Mechanism:** Independent auditing of the clinical and bioanalytical phases

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6-8 months (from screening through final PK clearance) **Onsite Visit Cadence:** Intensive inpatient monitoring post-dose followed by outpatient PK visits **Remote Monitoring Frequency:** N/A (primarily direct observation) **Checklist Review Interval:** Daily during the inpatient portion

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |